

ASCII MASTER FLOW – PANEL 1/12: End-to-end MHEWS rail

- 1 RISK KNOWLEDGE
- 2 MONITOR / OBSERVE / FORECAST
- 3 AUTHORITATIVE WARNING
- 4 DISSEMINATE -> 5 PREPARE / ACT -> 6 FEEDBACK

ASCII MASTER FLOW – PANEL 2/12: Risk-knowledge stack

HAZARD + EXPOSURE + VULNERABILITY + CAPACITY
SPATIAL LAYERS + HISTORICAL / REAL-TIME DATA
DECISION -> WHO WHERE WHEN WHAT ACTION
UPDATE -> COMMUNITY AND EVENT FEEDBACK

ASCII MASTER FLOW – PANEL 3/12: Observation-to-warning firewall

SENSOR / REPORT -> QUALITY CONTROL

MODEL + EXPERT ANALYSIS -> FORECAST / ASSESSMENT

COMPETENT AUTHORITY -> WARNING

PUBLIC ACTION -> NOT RAW DATA

ASCII MASTER FLOW – PANEL 4/12: Dissemination redundancy

CAP MESSAGE -> SMS / CELL / APP / WEB

CAP MESSAGE -> RADIO / TV / SIREN

LOCAL RELAY -> VOLUNTEER / PUBLIC ADDRESS

BACKUP -> POWER + NETWORK + MANUAL PATH

ASCII MASTER FLOW – PANEL 5/12: Warning-to-action test

RECEIVED? -> UNDERSTOOD?

TRUSTED? -> ACTION FEASIBLE?

ROUTE / TRANSPORT / SHELTER AVAILABLE?

RESULT -> EARLY ACTION OR LAST-MILE FAILURE

ASCII MASTER FLOW – PANEL 6/12: Agency ownership matrix

IMD -> WEATHER / CYCLONE

CWC -> RIVER FLOOD FORECAST

INCOIS -> TSUNAMI

ISRO / NRSC -> EARTH OBSERVATION / GIS

NDMA + STATE / LOCAL -> INTEGRATION AND ACTION

ASCII MASTER FLOW – PANEL 7/12: CAP and SACHET boundary

CAP -> structured message format

SACHET -> national alert portal / dissemination layer

SOLVES -> channel-format interoperability

DOES NOT SOLVE -> hazard ownership or local readiness

ASCII MASTER FLOW – PANEL 8/12: Geospatial decision cycle

SATELLITE / AERIAL / FIELD DATA

GIS LAYERS -> RISK MAP / ROUTE / SHELTER

EVENT IMAGE -> DAMAGE / ACCESS ASSESSMENT

RECOVERY MAP -> SAFER SITING

ASCII MASTER FLOW – PANEL 9/12: Sensor-hazard matching

DOPPLER RADAR -> WEATHER STRUCTURE

GAUGE -> RAIN / RIVER / SEA LEVEL

BUOY / BPR -> OCEAN RESPONSE

SEISMIC NETWORK -> EARTHQUAKE PARAMETERS

ASCII MASTER FLOW – PANEL 10/12: Emerging-technology safeguards

DRONE -> local imagery | airspace + safety

AI -> pattern / prioritisation | uncertainty + oversight

CROWD REPORT -> local signal | verification + provenance

DATA -> utility | privacy + minimisation

MUST REMEMBER: An end-to-end early-warning chain runs observation -> detection ->...

ASCII MASTER FLOW – PANEL 11/12: Failure and redundancy map

SENSOR FAIL | POWER FAIL | NETWORK FAIL

AUTHORITY DELAY | FORMAT MISMATCH

LANGUAGE / ACCESS BARRIER | NO TRANSPORT

CONTROL -> REDUNDANT TECHNICAL AND HUMAN PATHS

CLOSE DISTINCTION: Observation is not forecast; forecast is not watch; watch is not...

ASCII MASTER FLOW – PANEL 12/12: Technology answer spine

MAP HAZARD + LEAD TIME -> NAME COMPETENT AGENCY

TRACE DATA -> FORECAST -> WARNING -> DISSEMINATION

ADD ACCESSIBLE ACTION + FEEDBACK

QUALIFY UNCERTAINTY PRIVACY AND OUTCOME EVIDENCE

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Attribute the competent agency and...